Filip Ručka

Date of birth: 18/03/2004 | **Nationality:** Czech | **Email address:** filip.rucka@cern.ch | **Website:** https://filiprucka.github.io/web/ | **LinkedIn:** www.linkedin.com/in/filip-ručka-13688b329

ABOUT ME

I study Robotics at CTU in Prague and have a deep passion for fields surrounding robotics, such as machine learning, system control, and computer vision. I am currently working on my bachelor thesis at **CERN**, focusing on machine learning for the separation of the rare tH(bb) signal. I also work part-time at the **Czech Institute of Informatics, Robotics and Cybernetics** as a computer vision engineer. You can see more of my work on my website, where I have an in depth description of my robotics projects

WORK EXPERIENCE

CERN – PRAGUE, CZECHIA

MACHINE LEARNING RESEARCHER - BACHELOR THESIS – 01/02/2025 – CURRENT

- Conducting bachelor thesis research at **CERN** within the **ATLAS** experiment, focusing on improving separation of the rare tH(bb) signal from background processes in collision data.
- Developing and comparing machine learning models, including XGBoost and custom neural networks, using ROOT and TMVA.
- Performing hyperparameter optimization with Optuna and evaluating model performance through signal significance calculations.
- Collaborating with an international research team and presenting analysis results at conferences (Göttingen, 2025).

CZECH INSTITUTE OF INFORMATICS, ROBOTICS AND CYBERNETICS – PRAGUE, CZECHIA

COMPUTER VISION ENGINEER – 01/03/2025 – CURRENT

- Developed a complete computer vision pipeline in Python and OpenCV to automate crystal placement in a robotic cell used for preparing crystals for the neutron accelerator at **CERN**, in collaboration with the **Faculty of Mathematics and Physics at Charles University.**
- Implemented robust crystal boundary extraction using Sobel gradients combined with alpha-shape reconstruction to handle partial occlusions from the glass pipette.
- Designed a pipette-detection algorithm by sliding a window along the crystal boundary and applying least-squares circle fitting for accurate localization.
- Processed the target placement area by detecting the circular desk boundary and using image differencing to isolate the precise crystal-placement region.
- Determined crystal position and orientation by rotating an edge-based template and matching it against a gradient map of the target area.

FACULTY OF ELECTRICAL ENGINEERING, CZECH TECHNICAL UNIVERSITY IN PRAGUE – PRAGUE, CZECHIA

INTERN AT THE LAB OF COMPUTATIONAL ROBOTICS – 17/08/2024 – 26/11/2024

- Collected and processed data for reinforcement learning of robots navigating various terrains.
- involved in 3D printing components for robotic prototypes
- team work

EDUCATION AND TRAINING

24/09/2023 – CURRENT Prague, Czechia

BACHELOR OF ENGINEERING - CYBERNETICS AND ROBOTICS Faculty of Electrical Engineering, Czech Technical University in Prague

Courses: Robotics, Automatic control, Laboratory of robotics, Sensors and Measurement, Signals, systems and inference, Cybernetics and Artificial Intelligence, Mathematical optimization, Complex Analysis and Transformations

Projects:

Rotary Pendulum Modeling & Control:
Derived a mathematical model via system identification and implemented over 10 control strategies including **MPC**, **PID**, **DeadBeat** regulators. **Link:** https://filiprucka.github.io/web/ari.html

Autonomous Goal-Scoring Robot:
A TurtleBot leveraging **OpenCV**, and **SLAM** for robust computer vision, precise localization, and dynamic path planning to score goals autonomously with complex **obstacles**. **Link:** https://filiprucka.github.io/web/lary.html

Control of Mitsubishi RV-6S manipulator
Autonomous manipulation using computer vision, homography, and path planning on a 6-DoF Mitsubishi RV6S arm.
Link: https://filiprucka.github.io/web/rob.html

Website https://fel.cvut.cz | **Level in EQF** EQF level 6

01/01/2017 – 18/06/2023 Prague, Czechia
DUAL HIGH SCHOOL DIPLOMA: AUSTRIAN MATURA & CZECH MATURITA AND 1 YEAR EXCHANGE AT THE STIFTUNG THERESIANISCHE AKADEMIE IN VIENNA Austrian Highschool in Prague (Österreichisches Gymnasium Prag)

- Completed a six-year dual diploma program
- Earned both the Austrian Matura and the Czech Maturita
- Studied entirely in German for the last 4 years
- Selected through a competitive process from my entire grade and was awarded a full scholarship to attend the prestigious Theresianum in Vienna
- Lived at the academy for one year as part of the International Students Program

Website https://www.oegp.cz | **Level in EQF** EQF level 5

SKILLS

Technical skills

Robotics | Computer Vision | System Control | Machine Learning | C | Python | Git | OpenCV | MATLAB

CERTIFICATIONS

ETH Zürich, 18/10/2025
Autonomous Moboile Robots

Valid Certificate ID: ef2188fffc60475e92e3800232ee52ee

Mode of learning: Online

Link https://courses.edx.org/certificates/ef2188fffc60475e92e3800232ee52ee

ÖSD, 2021
German C1 ÖSD certification

Credential ID: ZC1j21m10350

Mode of learning: Presential

HONOURS AND AWARDS

Full 20 000 $ scholarship awardee - Theresian academy in Vienna

Selected through a competitive process from my entire grade

VOLUNTEERING

Project to Reduce Academic Failure Rates

Personally invited by the **Head** of the Department of Control Engineering and the **guarantor of the Master's program Cybernetics and Robotics** to participate in a project dedicated to reducing academic failure rates through a series of consultations.